

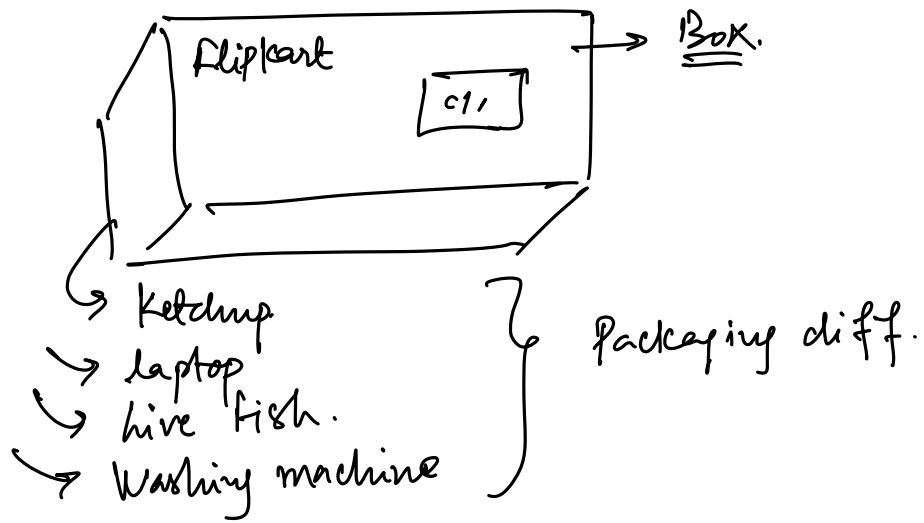
Agenda

① What are Generics

- Why Generics
- Generic Classes
- Generic methods & Static context } Interview trap.
- Bounded Types & Wildcards.

② Collections Framework. } Cheatsheet

Generics.



Daily Use Data Structures.

→ Array list, Maps, linked list, stacks, {Queues} etc
↓
Integers, Strings, Student.

What type in Java can hold anything? → object =

```
def get-student-info():  
    return ("Rahul", 95);
```

tuple → holds 2 values.

```
name, score = get-student-info()
```

→ The core problem

- * No type safety.
- * Manual type casting :- (String) getValue(); → Not good.
- * Runtime bombs. :- Error will only come to surface when the code runs and some edge cases are hit.

Without Generics (Object Type)

```
Pair data = new Pair("Rahul", 95);  
data.setFirst(42); → Compiles.  
Crashes at runtime
```

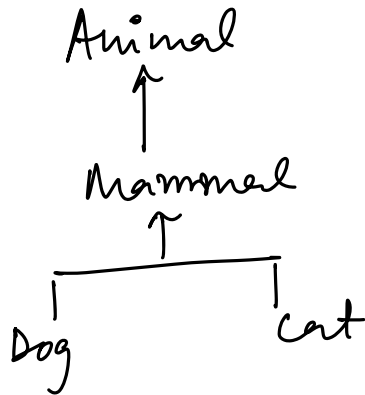
Casting is required

With Generics:-

```
Pair<String, Integer> data = new Pair<>("Rahul", 95);  
data.setFirst(42);  
→ compile error.
```

No casting required.

Bounded Types & Wildcards.



Dog rex = new Dog();

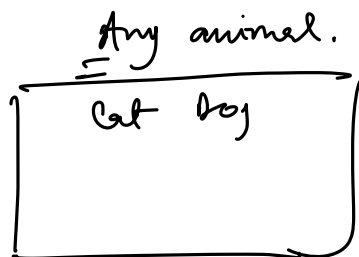
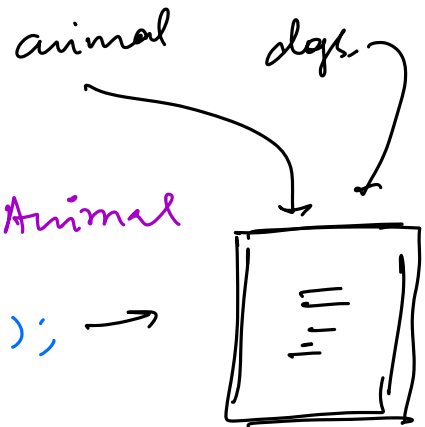
Animal a = rex; → Yes, Dog **IS-A** Animal

List<Dog> dogs = new ArrayList<>();
dogs.add(new Dog());

List<Animal> animals = dogs; X Wrong → compile time error.

animals.add(new Cat());

Dog d = dogs.get(dogs.size()-1); → We just pulled out a Cat from a list of dogs.



Dog extends Animal. $\rightarrow$ Dog is-A animal

list <Dog> extends list <Animal> X

$\rightarrow$ list <Dog> is NOT
an list <Animal>

* Java generics are invariant with respect to their
type parameters.

? extends T $\rightarrow$ read-only

You can take thing out
But cannot put more thing IN because you might
put the wrong subtype (N).

? super T $\rightarrow$ when you write.

P E C S
 ~~~~~  
 Producer extends  
 ~~~~~  
 Consumer Super.

Direction	Wildcard	Meaning
Reading FROM (producing data)	? extends T	Give me anything that IS-A T
Writing INTO (consuming data)	? super T	Give me a container that can hold T.

? extends Animal → Read FROM the list → 'Producer' of Animal values.

? super Dog → WRITE to the list → 'Consumer' of Dog values.